

First results on the estimation of cointegrating relations for state space models in the SPECM framework when exogenous variables are included

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- Consider the modelling of vector time series, which is assumed to be co-integrated.
- Specification and estimation is easier, if the model can be partitioned into a number of sub-models which can be specified and estimated separately.
- In general, this is not advisable, as statistical efficiency is only obtained for the joint model.
- However, under assumptions of (strong) exogeneity, no loss in accuracy is incurred.
- For VAR models in the context of co-integrated systems Johansen (1997) and Pradel and Rault (2003, OBES) derive conditions such that the estimation of co-integrating relations in 'partial systems' can be done efficiently in VAR/VECM models.
- This talk provides first results for an analogous investigation in the state space framework.

1. Problem Description



- Consider two processes $(y_t), y_t \in \mathbb{R}^s$ and $(z_t), z_t \in \mathbb{R}^p$.
- We want to model (y_t) endogenously, while (z_t) is to be modelled as exogenous.
- For the presentation we first assume that (z_t) has a state space representation:

$$\begin{aligned}z_t &= C_z x_{t,z} + \varepsilon_{t,z}, \\x_{t+1,z} &= A_z x_{t,z} + K_z \varepsilon_{t,z}.\end{aligned}$$

- Here $A_z - K_z C_z$ is assumed to be stable such that (z_t) has a VAR(∞) representation:

$$z_t = \sum_{j=1}^{\infty} \Phi_{j,z} z_{t-j} + \varepsilon_{t,z}.$$

- A_z may contain unit eigenvalues such that it has a Jordan normal form as:

$$A_z = T_A \begin{pmatrix} I_{c_z} & 0 \\ 0 & A_{z,\bullet} \end{pmatrix} T_A^{-1}$$

where $A_{z,\bullet}$ is stable (has eigenvalues smaller than 1 in modulus).

- The process (y_t) is assumed to possess a state space representation of the following form:

$$\begin{aligned} y_t &= Cx_t + Dz_t + \varepsilon_t, \\ x_{t+1} &= Ax_t + Bz_t + K\varepsilon_t. \end{aligned}$$

- We assume invertibility such that $A - KC$ is stable.
- Again A may have unit eigenvalues such that the basis of the state space can be chosen to obtain the following representation:

$$\begin{aligned} y_t &= C_c x_{t,c} & + C_\bullet x_{t,\bullet} & + Dz_t & + \varepsilon_t, \\ x_{t+1,c} &= x_{t,c} & & + B_c z_t & + K_c \varepsilon_t & (*) \\ x_{t+1,\bullet} &= & A_\bullet x_{t,\bullet} & + B_\bullet z_t & + K_\bullet \varepsilon_t \end{aligned}$$

- $x_{t,c} \in \mathbb{R}^c$ denotes the components of the state for (y_t) where summation occurs.

The joint process thus has the representation:

$$\begin{pmatrix} I & -D \\ 0 & I \end{pmatrix} \begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} C & 0 \\ 0 & C_z \end{pmatrix} \begin{pmatrix} x_t \\ x_{t,z} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ \varepsilon_{t,z} \end{pmatrix},$$

$$\begin{pmatrix} x_{t+1} \\ x_{t+1,z} \end{pmatrix} = \begin{pmatrix} \underline{A} & 0 \\ 0 & \underline{A}_z \end{pmatrix} \begin{pmatrix} x_t \\ x_{t,z} \end{pmatrix} + \begin{pmatrix} K & B - KD \\ 0 & K_z \end{pmatrix} \begin{pmatrix} y_t \\ z_t \end{pmatrix}$$

- This is an innovation state space representation only if the joint vector $(\varepsilon'_t, \varepsilon'_{rt,z})'$ is WN.
- Since Dz_t is included in the model, uncorrelatedness of ε_t and $\varepsilon_{t,z}$ can be assumed wrlg.
- However, $\varepsilon_{t,z}$ not necessarily is uncorrelated with $\varepsilon_{t-j}, j > 0$.

Questions:

- Under what assumptions is the joint representation a state space representation?
- What are the properties of (y_t) then?
- What restrictions are needed for strong exogeneity of (z_t) ?
- Under strong exogeneity conditions, how can one specify the integer c of the number of common trends generated within the system for (y_t) ?

- The joint representation for WN $(\varepsilon'_t \varepsilon'_{t,z})'$ implies the VAR(∞):

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \tilde{u}_t + \sum_{j=1}^{\infty} \begin{pmatrix} \Phi_{yy,j} & \Phi_{yz,j} \\ 0 & \Phi_{zz,j} \end{pmatrix} \begin{pmatrix} y_{t-j} \\ z_{t-j} \end{pmatrix}$$

- This implies that y_t does not Granger-cause z_t .
- If, on the contrary, $\varepsilon_{t,z}$ can be predicted using $\varepsilon_{t-j}, j > 0$, y_t does Granger-cause z_t .
- This can be tested in a joint state space model, as the representation above includes three zero matrices (two blocks in C , one in the Kalman gain; one block in C additionally is restricted by the model given above).

- From now on, we assume that (y_t) does not Granger-cause (z_t) and hence the joint representation is a state space representation in innovation form.
- Therefore the joint process $([y_t', z_t']')$ has a $\text{VAR}(\infty)$ representation, wherein (y_t) does not Granger-cause (z_t) : Using $\underline{A} = A - KC, \underline{A}_z = A_z - K_z C_z$

$$\begin{pmatrix} I & -D \\ 0 & I \end{pmatrix} \begin{pmatrix} y_t \\ z_t \end{pmatrix} = \sum_{j=1}^{\infty} \begin{pmatrix} C \underline{A}^{j-1} K & C \underline{A}^{j-1} (B - KD) \\ 0 & C_z \underline{A}_z^{j-1} K_z \end{pmatrix} \begin{pmatrix} y_{t-j} \\ z_{t-j} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ \varepsilon_{t,z} \end{pmatrix}$$

3. Properties of y_t



- The common trends in (z_t) are given as $C_{z,1}x_{t,z,1}$ which are hence $I(1)$.
- Then $z_t = C_{z,1}x_{t,z,1} + C_{z,\bullet}x_{t,z,\bullet} + \varepsilon_{t,z}$ such that in the state equation for (y_t) the entries $B_c C_{z,1}x_{t,z,1}$ are doubly integrated.
- Therefore $B_c C_{z,1} = 0$ is a necessary condition for (y_t) to be $I(1)$ rather than $I(2)$.
- Also the state $(x_{t,\bullet})$ potentially is not stationary, if $B_\bullet C_{z,1} \neq 0$.
- The impact of the common trends in z_t for y_t equal

$$k_\bullet(1)C_{z,1}x_{t,z,1} = (D_z + C_\bullet(I - A_\bullet)^{-1}B_\bullet)C_{z,1}x_{t,z,1}.$$

- The common trends in (y_t) thus equal

$$k_\bullet(1)C_{z,1}x_{t,z,1} + C_c \sum_{j=1}^{t-1} (B_c z_j + K_c \varepsilon_j).$$

The State sPace Error correction model (SPECM) representation for the joint $I(1)$ process equals:

$$\begin{pmatrix} I & -D \\ 0 & I \end{pmatrix} \begin{pmatrix} \Delta y_t \\ \Delta z_t \end{pmatrix} = \Pi \begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} - \begin{pmatrix} C\underline{A}(I - \underline{A})^{-1} & 0 \\ 0 & C_z\underline{A}_z(I - \underline{A}_z)^{-1} \end{pmatrix} \begin{pmatrix} \tilde{x}_t \\ \tilde{x}_{t,z} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ \varepsilon_{t,z} \end{pmatrix},$$

$$\begin{pmatrix} \tilde{x}_{t+1} \\ \tilde{x}_{t+1,z} \end{pmatrix} = \begin{pmatrix} \underline{A} & 0 \\ 0 & \underline{A}_z \end{pmatrix} \begin{pmatrix} \tilde{x}_t \\ \tilde{x}_{t,z} \end{pmatrix} + \begin{pmatrix} K & B - KD \\ 0 & K_z \end{pmatrix} \begin{pmatrix} \Delta y_t \\ \Delta z_t \end{pmatrix}$$

Here

$$\begin{aligned} \Pi &= \begin{pmatrix} -I & D \\ 0 & -I \end{pmatrix} + \begin{pmatrix} C(I - \underline{A})^{-1} & 0 \\ 0 & C_z(I - \underline{A}_z)^{-1} \end{pmatrix} \begin{pmatrix} K & B - KD \\ 0 & K_z \end{pmatrix} \\ &= \begin{pmatrix} \alpha\beta' & D + C(I - \underline{A})^{-1}(B - KD) \\ 0 & \alpha_z\beta_z' \end{pmatrix} \end{aligned}$$

Thus the first equation is given by

$$\Delta y_t - D\Delta z_t = \alpha\beta' y_{t-1} + \Pi_{1,2} z_{t-1} - C\underline{A}(I - \underline{A})^{-1} \tilde{x}_t + \varepsilon_t.$$

- Key to exogeneity properties then is

$$\begin{aligned}\Pi_{1,2} &= C(I - \underline{A})^{-1}B + D - C(I - \underline{A})^{-1}KD \\ &= C(I - \underline{A})^{-1}B - \alpha\beta'D.\end{aligned}$$

- If $C(I - \underline{A})^{-1}B = \alpha\tilde{\beta}'$ then $\Pi_{1,2} = \alpha\beta'_*$ such that

$$\begin{aligned}\Pi &= \begin{pmatrix} \alpha & 0 \\ 0 & \alpha_z \end{pmatrix} \begin{pmatrix} \beta' & \beta'_* \\ 0 & \beta'_z \end{pmatrix}, \\ \Delta y_t - D\Delta z_t &= \alpha(\beta' y_{t-1} + \beta'_* z_{t-1}) - C\underline{A}(I - \underline{A})^{-1}\tilde{x}_t + \varepsilon_t.\end{aligned}$$

- Then the equation for Δy_t does not contain parameters featuring in the model for z_t (=sequential cut in the parameters).

Equivalence to $B_c = 0$

- Since (A, K, C) is assumed to be in canonical form, we get $C_c K_c = \beta_{\perp} (\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp}$ (follows from Granger representation theorem), we get $K_c = T_K \alpha'_{\perp}$.
- Then

$$\begin{aligned} I - \underline{A} &= I - A + KC = \begin{pmatrix} 0 & 0 \\ 0 & I - A_{\bullet} \end{pmatrix} + KC \Rightarrow [I, 0](I - \underline{A}) = K_c C \\ \Rightarrow [I, 0] &= K_c C(I - \underline{A})^{-1} \\ B_c &= [I, 0]B = K_c C(I - \underline{A})^{-1}B = T_K \alpha'_{\perp} C(I - \underline{A})^{-1}B. \end{aligned}$$

- Thus, if $C(I - \underline{A})^{-1}B = \alpha \tilde{\beta}'$ we get $B_c = 0$.
- Conversely, if $B_c = 0$ it follows that $\alpha'_{\perp} C(I - \underline{A})^{-1}B = 0$ showing the equivalence.

Theorem (Strong Exogeneity)

In the model () there exists a sequential cut in the parameters for the marginal model and for the conditional model, if and only if $B_c = 0$.*

- In that case the common trends in (y_t) are given as

$$\sum_{j=0}^{t-1} C_c K_c \varepsilon_j + k_{\bullet}(1) \sum_{j=0}^{t-1} C_{z,1} K_{z,1} \varepsilon_{t,z}.$$

- This clarifies the co-integrating relations.
- Note that under strong exogeneity, the marginal model for (z_t) is not needed in the estimation of the conditional model. Thus the assumption of joint state space representation could also be dropped.

5. Testing for Strong Exogeneity



- Strong Exogeneity involves two separate properties: Granger non-causality of (y_t) for (z_t) and the splitting property for the parameter sets.
- Granger non-causality can be tested in the current model, as Granger causality implies a restriction of the parameter set.
- Given Granger non-causality, the sequential cut property for the parameters hinges on $B_c = 0$, which clearly is a linear restriction for the conditional model.
- Details for the two tests, for example their asymptotic distributions are future research.

- The SPEXM then is exactly the augmented model of Rahbek and Mosconi (1999) where their stationary exogenous process equals $\Delta(z_t)$.
- Their augmentation then is exactly the SPEXM setting (they deal with a VECM, however).
- The asymptotic distribution of the trace test then depends on the co-integrating rank for (z_t) , see Rahbek and Mosconi (1999), Remark 3. Beside this rank there are no other nuisance parameters.
- The dependence on the cointegrating rank can be eliminated by using the maximum over all ranks.

Not strongly exogenous: consequences

- If strong exogeneity does not hold and $B_c \neq 0$, the stationary part of z_t also adds to the common trends in (y_t) .
- The model does not admit a sequential cut any more, since – for example – $B_{i,c}C_{z,1} = 0$ needs to be imposed to exclude $I(2)$ systems in the model.
- Also, shocks of (z_t) may have permanent effects on (y_t) and only short term impacts on (z_t) .
- Thus in this situation long run properties of (y_t) require modelling of (z_t) .